CMT 316:	HUMAN COMPUTER INTERFACE	**3 CREDIT HOURS**
PREREQUISITE:		

Purpose:

- To introduce students to concepts, techniques, skills and methodologies for designing usable, user friendly, safe, interactive, efficient, effective and artefacts.

Leaning outcomes

- By the end of the module, students should be able to:

(a) Appreciate the impact of user friendly systems/artefacts in our society.

(b) Gain critical understanding of the impact of modern user centered interface/interaction

(c) design theories and concepts on product and system design and development;

(d) Review current emerging trends revolving around Human centered systems

(e) Carry out a self-directed research project

Course Description

1. Introduction to HCI, motivation to HCI, technology changes and different HCI needs, goals of HCI, disciplines contributing to HCI, HCI principles and concepts
2. Cognitive Psychology; Introduction, Contribution/importance of Psychology to HCI, core cognitive aspects Human Brain, Psychological Models used in HCI - The Human Processor Model, Memory - STM & LTM, techniques of improving STM, George Millar theory, Design & context
3. Interaction & emerging computing paradigm shifts; introduction, terms of interaction, interaction i.e. GOMS, Donald execution model and Abowd and Beale Model,
4. Emerging computing paradigms e.g. cloud computing, mobile computing, agent based computing, sensor based computing, ubiquitous computing
5. User centered design; User profile analysis, Requirement analysis, Task Description and Analysis
6. Usability Engineering; goals, principles, guidelines and standards
7. Mobile interaction design: introductions, approaches of mobile development; hybrid, native, web based, challenges facing mobile programmers & solutions
8. Internationalization Vs Localization; introduction, challenges & solutions being offered by toolkits
9. Designing for accessibility; introduction, categories of people with special needs in the society, how to incorporate people with special needs
10. Prototyping and Evaluation; introduction, prototyping fidelity, approaches of prototyping, evaluation techniques
11. Technology Acceptance Model (TAM)

Teaching Methodologies:

Lectures, demonstrations, group/class discussions and practical exercises.

Instructional Materials/Equipment:

Computers, projectors, whiteboards and writing materials.

1. **Course Assessment:**

2. **Assessment Strategy:**	**Weighting**
1. CATS	10%
2. Semester based project	30%
3. End-of-semester examination	60%
Total	**100%**

Textbooks for the course

1. *Designing User Experience: A guide to HCI, UX and interaction design (4th Edition) 4th Edition* (4th ed.). (January 19, 2019). Pearson.
2. Michael Filimowicz, V. T. (July 3, 2018). *New Directions in Third Wave Human-Computer Interaction: Volume 1* (1st ed.). Springer.
3. Sharp, H., Jennifer Preece , & Yvonne Rogers . (May 29, 2019). *Interaction Design: Beyond Human-Computer Interaction* (5th ed.). Wiley.